
Software Requirements Specification

For
SME – GPT
v1.2

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2026.02.10

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Revision History

Name	Date	Reason For Changes	Version
Requirement analysis	5.12.2025	Initial draft created	1.0
Requirement Analysis 2.0	25.12.2025	Added functional and nonfunctional requirements	1.1
SRS Document	10.02.2026	Updated based on supervisor feedback	1.2

1. Introduction

1.1. Purpose

This Software Requirements Specification (SRS) document specifies the software requirements for the project “SME - GPT: Explainable AI System for Sinhala & English Financial Document Understanding” (Version 1.2). The SRS defines the functional features, quality constraints, and external interface needs of the proposed system. It describes what the system must do to support Sri Lankan SMEs in extracting and reasoning over financial documents written in Sinhala and English.

This SRS is written to guide the design, development, testing, and evaluation of the system. It provides a structured reference for academic supervision, team coordination, and validation planning. The scope described in this document covers the full software system in its initial research prototype release, focusing on document upload, OCR extraction, layout understanding, structured field extraction, RAG-based question answering, and explainability.

1.2. Document Conventions

This document follows IEEE-style conventions and the given university template structure. Requirements are expressed using “shall” for mandatory requirements, “should” for recommended requirements “may” for optional capabilities.

Functional requirements are labeled FR-xx and non-functional requirements are labeled NFR-xx exactly as provided. Each requirement includes a short rationale (“Why”) to clarify the business and technical motivation. Unless stated otherwise, priorities for detailed requirements are assumed to be inherited from the parent feature priority.

1.3. Intended Audience and Reading Suggestions

This SRS is intended for:

- Developers (team members): implement features exactly as specified in Section 3
- Academic supervisor / mentor: evaluate completeness, feasibility, and correctness of scope
- Testers / evaluators: derive test cases from requirements (Sections 3 and 5)
- Documentation writers: prepare user manuals and help content (Section 2.6, Section 4.1)
- SME stakeholders: understand system value, limits, and expected user experience

Reading order recommendation:

- Start with Section 1 & 2 for overview and scope
- Read Section 3 for detailed functional requirements
- Read Section 4 for interface expectations
- Read Section 5 for quality attributes and constraints
- Check appendices for diagrams, glossary, and open issues

1.4. Project Scope

The project aims to develop an AI system that processes financial documents (invoices, purchase orders, delivery notes) in PDF and image formats. The system will support Sinhala and English OCR, extract key financial fields and line-item tables, store structured outputs, and allow users to ask natural language questions. The system will provide explainable outputs by showing provenance (page number, bounding boxes, and source text) so SMEs can trust and verify results.

Out of scope for Version 1.2: handwritten recognition at enterprise-level accuracy, full accounting automation, payment processing, inventory management, and deep ERP integration.

1.5. References

- IEEE Recommended Practice for Software Requirements Specifications (IEEE 830 / ISO/IEC/IEEE 29148 guidance)
- OCR Engine documentation (Surya / alternatives)
- FastAPI documentation (backend)
- React documentation (frontend)
- PostgreSQL and vector database documentation
- Research papers on Layout Analysis, RAG, and Explainable AI

2. Overall Description

2.1. Product Perspective

Multimodal Doc-GPT for SMEs is a new, standalone application designed to improve SME financial document handling. It does not replace existing accounting software; instead, it works as a supporting intelligence layer that extracts, organizes, and answers questions over documents. The system sits between the user and their document archive and converts unstructured content into a structured and searchable knowledge base. At a high level, the system contains:

- 1) UI for upload, viewing, and Q&A
- 2) Backend APIs for orchestration and access control
- 3) OCR service for Sinhala/English text extraction
- 4) Layout & extraction service for tables and key fields
- 5) Storage layer (documents, OCR, metadata, logs)
- 6) Vector indexing for retrieval
- 7) RAG engine for grounded answers + calculator for correct math
- 8) Explainability layer for provenance and highlights

2.2. Product Features

Key features include:

- 1) Document upload and preprocessing
- 2) Sinhala + English OCR with bounding boxes and confidence
- 3) Layout detection for tables and structured blocks
- 4) Extraction of key invoice fields and line items
- 5) Indexing and semantic retrieval via vector embeddings
- 6) Natural language question answering using RAG
- 7) Deterministic arithmetic checking for totals
- 8) Explainability and provenance visualization in UI
- 9) Security: RBAC, encryption, audit logs

2.3. User Classes and Characteristics

- 1) Admin:
manages users, roles, access policy; high privilege, moderate technical skill.
- 2) Accountant/Finance user:
uploads documents, verifies extracted data, runs queries; domain expert, medium technical skill.

3) SME Owner/Manager:

asks high-level questions (monthly totals, supplier spend); low technical skill; needs very simple UX.

4) Auditor/Reviewer:

validates extracted values and evidence trails; needs strong provenance, logs, and transparency.

2.4. Operating Environment

- 1) Frontend: Modern browsers (Chrome/Edge/Firefox), desktop & mobile
- 2) Backend: Linux/Windows server environment
- 3) Deployment: Docker containers recommended for portability
- 4) Storage: Local disk / cloud object storage (depending on deployment)
- 5) Internet: Required if using hosted LLM APIs; optional if local LLM configured

2.5. Design and Implementation Constraints

- 1) Sinhala OCR quality can vary due to fonts, low-quality photos, scanning noise
- 2) SMEs may have limited hardware; system must be efficient
- 3) Sensitive financial data requires strong security practices
- 4) Must support bilingual UI and bilingual query handling
- 5) Modular design required so OCR/LLM components can be replaced/upgraded
- 6) Must ensure answers remain grounded (avoid hallucinations)

2.6. User Documentation

The software will provide:

- User manual (Sinhala + English)
- Quick-start guide for upload, extraction review, and Q&A
- In-app tooltips and help icons
- Error-message guidance for common issues (blurred images, corrupted PDF, etc.)

2.7. Assumptions and Dependencies

Assumptions:

- Documents are mostly printed and readable (not extremely damaged)
- Users have basic internet access (if using cloud LLM)
- SMEs will provide representative invoice/PO samples for evaluation

Dependencies:

- OCR engine availability and Sinhala language capability
- Vector database availability
- LLM availability (API or local model)
- PDF rendering libraries for page conversion
- Browser compatibility for overlay/highlight rendering

3. System Features

3.1. Document Input & Preprocessing

3.1.1. Description and Priority:

This feature enables the system to accept and prepare documents for OCR and extraction. Because SMEs commonly upload phone photos and scanned PDFs, preprocessing is essential to improve OCR quality and reduce errors. This feature is High priority because it is the entry point for the entire pipeline; if ingestion fails, no other feature can function.

3.1.2. Stimulus/Response Sequences:

- User uploads PDF/image → system validates → confirms upload OR rejects with reason
- If PDF → system converts pages to images → stores page images
- System preprocesses images (deskew/denoise/crop) → forwards to OCR service
- If document unreadable → system stops pipeline and displays clear error

3.1.3. Functional Requirements:

FR-01: The system shall accept PDFs and images (JPG/PNG).

Why: Users upload different formats; SMEs often use phone photos.

FR-02: Convert PDF pages into high-quality images (300 DPI).

Why: A unified image pipeline improves OCR accuracy.

FR-03: Perform preprocessing (deskew, denoise, crop).

Why: Many photos are tilted or blurry; cleaning improves text recognition.

FR-04: Reject corrupted/unreadable documents with a clear error message.

Why: Prevents system failure and guides the user to fix the input.

3.2. OCR (Sinhala + English)

3.2.1. Description and Priority:

This feature extracts bilingual text from documents. OCR must output not only text but also positional metadata for explainability and auditability. It is High priority because downstream extraction and RAG depend on accurate OCR output.

3.2.2. Stimulus/Response Sequences:

- Preprocessed images arrive → OCR runs Sinhala+English extraction
- OCR returns detected text segments with bounding boxes and confidence scores
- System stores OCR output and links it with document/page metadata
- If OCR fails → system retries or reports failure reason

3.2.3. Functional Requirements:

FR-05: Extract text in Sinhala & English.

Why: SMEs require bilingual extraction for local documents.

FR-06: Output bounding boxes for every text segment.

Why: Required for explanation and provenance (where data came from).

FR-07: Store OCR confidence levels.

Why: Helps detect questionable text and debug extraction issues.

FR-08: Support pluggable OCR engines.

Why: Enables improvements and future Sinhala accuracy tuning.

3.3. Layout Detection & Data Extraction

3.3.1. Description and Priority:

Invoices and POs contain tables, headers, totals, and mixed formats. Layout detection organizes OCR results into meaningful structure to support accurate extraction. This is High priority because wrong layout parsing leads directly to wrong totals and wrong field values.

3.3.2. Stimulus/Response Sequences:

- OCR output received → system detects blocks/regions and table areas
- Key-value fields extracted (vendor, date, invoice no.)
- Line items extracted as structured rows/columns
- Each extracted field stored with page + bounding box
- If structure not detected → system falls back to heuristic extraction and flags uncertain

3.3.3. Functional Requirements:

FR-09: Detect document structure (tables, headers, blocks).

Why: Invoices aren't plain text; structure is required for organized extraction.

FR-10: Extract key fields (vendor, invoice number, dates, totals).

Why: These are the main fields SMEs need daily.

FR-11: Extract line-item tables with row/column structure.

Why: Line items drive totals and analytics (qty, unit price, etc.).

FR-12: Support multi-page extraction.

Why: Some invoices/POs span multiple pages.

FR-13: Store page number and bounding box for each extracted field.

Why: Enables explainability and auditability.

3.4. Indexing & Vector Retrieval

3.4.1. Description and Priority:

This feature converts extracted content into vector embeddings and supports semantic retrieval. It is High priority for the Q&A capability because RAG depends on retrieving relevant chunks reliably.

3.4.2. Stimulus/Response Sequences:

- Extracted text/fields are chunked → embedding model generates vectors
- Vectors stored in vector DB with metadata links to source document/page/bbox
- User asks question → retrieval returns top-k relevant chunks + provenance
- Evidence passed to the LLM for grounded answer generation

3.4.3. Functional Requirements:

FR-14: Convert extracted content into vector embeddings.

Why: Enables semantic search and RAG-based answers.

FR-15: Store embeddings in a vector database.

Why: Supports fast retrieval at scale.

FR-16: Support semantic retrieval for answering user queries.

Why: Users ask natural language questions, not keywords.

FR-17: Return provenance metadata with search results.

Why: Ensures transparency of retrieved evidence.

3.5. LLM-Based Question Answering

3.5.1. Description and Priority:

This feature provides conversational Q&A while ensuring answers are grounded in user documents. It is High priority because it represents the core “assistant” capability of the system. The system must avoid hallucinations and must compute arithmetic correctly.

3.5.2. Stimulus/Response Sequences:

- User submits question (Sinhala/English)
- System retrieves relevant evidence via vector search
- RAG prompt built using evidence only
- LLM generates answer → calculator verifies numeric computations
- System returns final answer + provenance references
- If evidence insufficient → system refuses or asks for clarification safely

3.5.3. Functional Requirements:

FR-18: Accept natural-language questions (Sinhala/English).

Why: Improves accessibility for local users.

FR-19: Use RAG pipeline to retrieve context before answering.

Why: Prevents hallucinations and keeps answers evidence-based.

FR-20: Use a calculator function for arithmetic.

Why: LLMs make math mistakes; calculator ensures correctness.

FR-21: Support multi-document reasoning (sum of all invoices etc.).

Why: SMEs often need totals across many documents.

FR-22: Only answer when provenance is available.

Why: Ensures trustworthiness and transparency.

3.6. Explainability & Provenance Tracking

3.6.1. Description and Priority:

Explainability is essential in financial contexts. Users must verify “where the answer came from” and “how totals were computed.” This is High priority and a distinguishing requirement of the project.

3.6.2. Stimulus/Response Sequences:

- Extraction occurs → system stores provenance metadata
- User views extracted fields → clicking shows origin on document image
- User asks question → answer returns highlighted evidence + calculation trace
- If OCR confidence low → UI warns user and suggests verification

3.6.3. Functional Requirements:

FR-23: Store full provenance: bounding box, page, raw text, model version.

Why: Critical for traceability and auditing.

FR-24: Highlight exact source text in the UI.

Why: Users must verify values and trust the system.

FR-25: Show derivation steps for aggregated answers.

Why: Users must understand how totals were computed.

FR-26: Allow user to click values to see origin.

Why: Interactive explainability improves usability and trust.

3.7. User Interface

3.7.1. Description and Priority:

The system must be easy for non-technical SME users. It must present documents, overlays, extracted results, and chat Q&A in a simple and bilingual manner. This is High priority.

3.7.2. Stimulus/Response Sequences:

- User logs in → dashboard shows documents and statuses
- User uploads → sees progress and processing state
- User opens document viewer → sees highlights and extracted fields
- User asks question → sees answer + evidence + trace
- Errors displayed clearly and politely

3.7.3. Functional Requirements:

FR-27: Provide document viewer with overlays.

Why: Users need to visually cross-check extracted information.

FR-28: Provide bilingual UI (Sinhala/English).

Why: Essential for local adoption.

FR-29: Provide clear errors and status updates.

Why: Reduces confusion and improves completion rate.

3.8. Security & Access Control

3.8.1. Description and Priority:

Financial documents contain sensitive values, supplier data, and sometimes personal information. Security must be designed in from the start. This is High priority.

3.8.2. Stimulus/Response Sequences:

- User logs in → receives authenticated session
- Access to documents limited by role and tenant
- Sensitive operations create audit log entries
- All communications encrypted; data encrypted at rest

3.8.3. Functional Requirements:

FR-30: Use TLS for all communication.

Why: Protects data in transit.

FR-31: Encrypt stored data using AES-256.

Why: Protects sensitive documents at rest.

FR-32: Use Role-Based Access Control (RBAC).

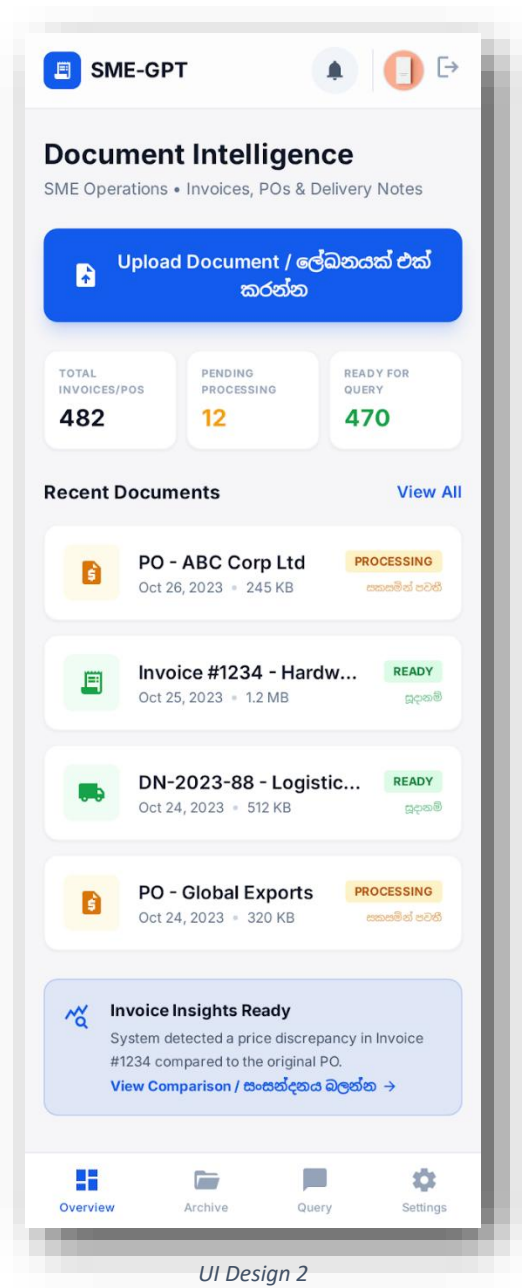
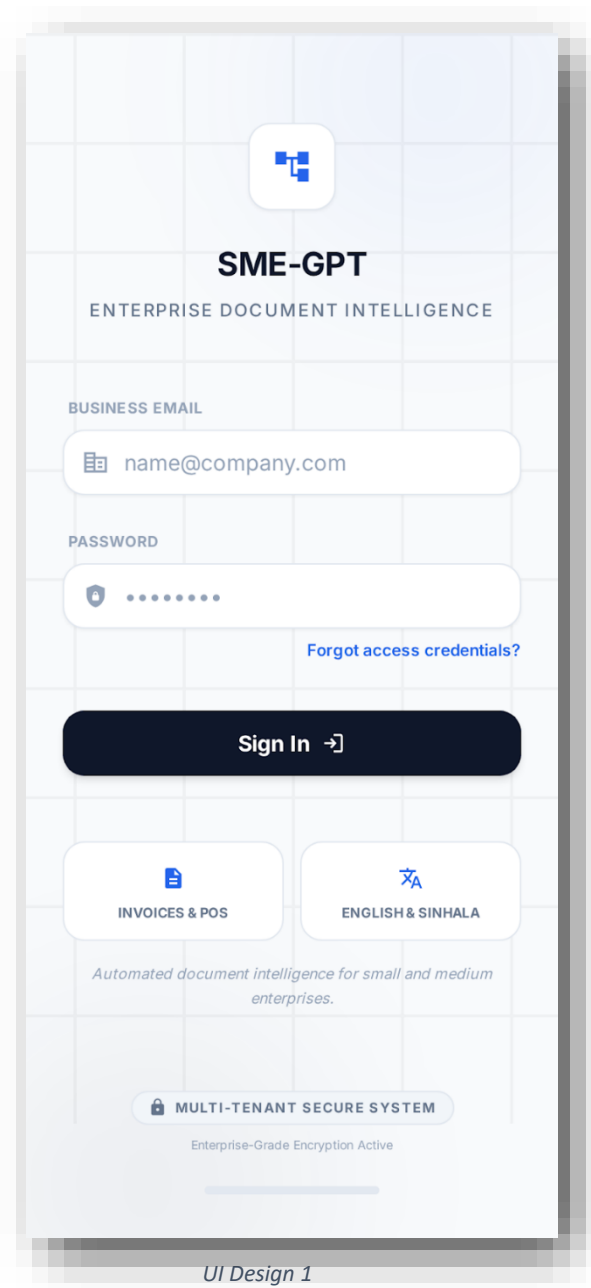
Why: Prevents unauthorized access.

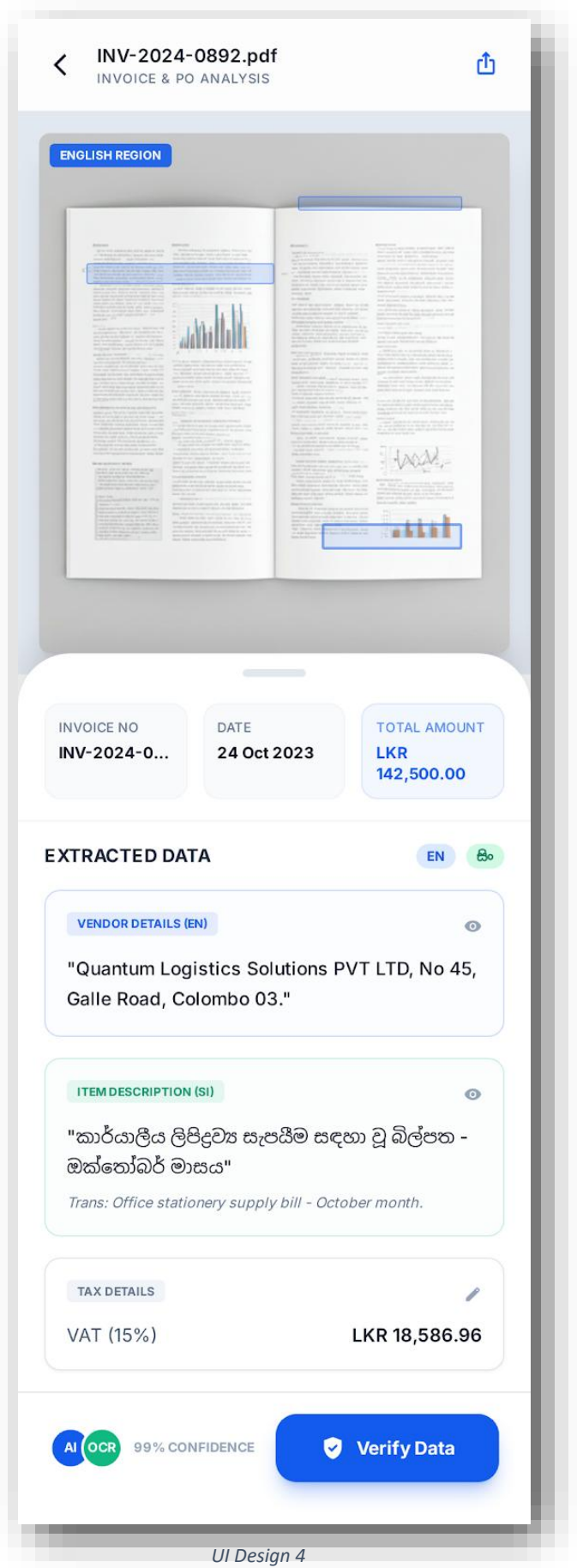
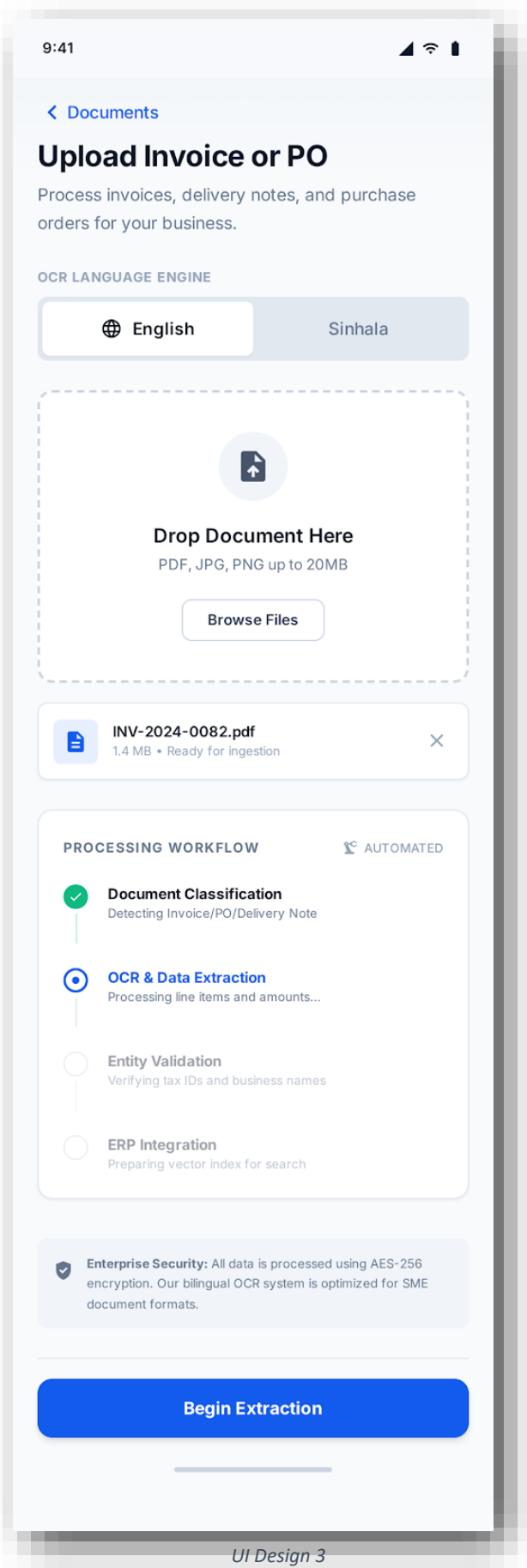
FR-33: Maintain audit logs for all actions.

Why: Required for accountability and compliance.

4. External Interface Requirements

4.1. User Interfaces





9:41


[< Back](#)

DOCUMENT AI

Query Invoices & POs

Ask questions about your invoices, delivery notes, or purchase orders in English or Sinhala.

What is the total amount in this invoice?

කරුණාකර මෙම බිල්පතේ මුළු මුදල කියන්න?



Auto-Detection active

Insight: You can ask about VAT details, vendor names, or specific line item quantities from multiple documents at once.

Process Query



OCR



NLP



XAI

UI Design 5

9:41


 INSIGHTS & ANALYSIS
Invoice Verification


QUERY

"Check if the unit price for 20kg Cement bags matches PO-882 and highlight discrepancies in Sinhala."

AI BUSINESS INSIGHT

98% ACCURACY

A discrepancy was found in **Invoice #INV-2024-012**. The unit price for Portland Cement (20kg) is listed at **LKR 2,450.00**, which is **8.8% higher** than the agreed price in **PO-882** (LKR 2,250.00).

සිංහල පරිවර්තනය

මෙම ඉන්වොයිසියේ සඳහන් සීමෙන්ති (20kg) ඒකක මිල, මිලදී ගැනීමේ ඇණවුමට (PO-882) වඩා වැඩි වේ. ඔබ ගෙවිය යුතු මුදල රු. 2,250.00 වන නමුත් මෙහි රු. 2,450.00 ක් ලෙස සටහන්ව ඇත.

ADJUST TOTAL

NOTIFY SUPPLIER

EXPORT

FLAG FOR REVIEW

ROW 12

OCR Confidence: 99.8%

Item: Portland Cement (20kg Bag)

2,450.00

PO-882 (Contract)

[GOTO PO](#)

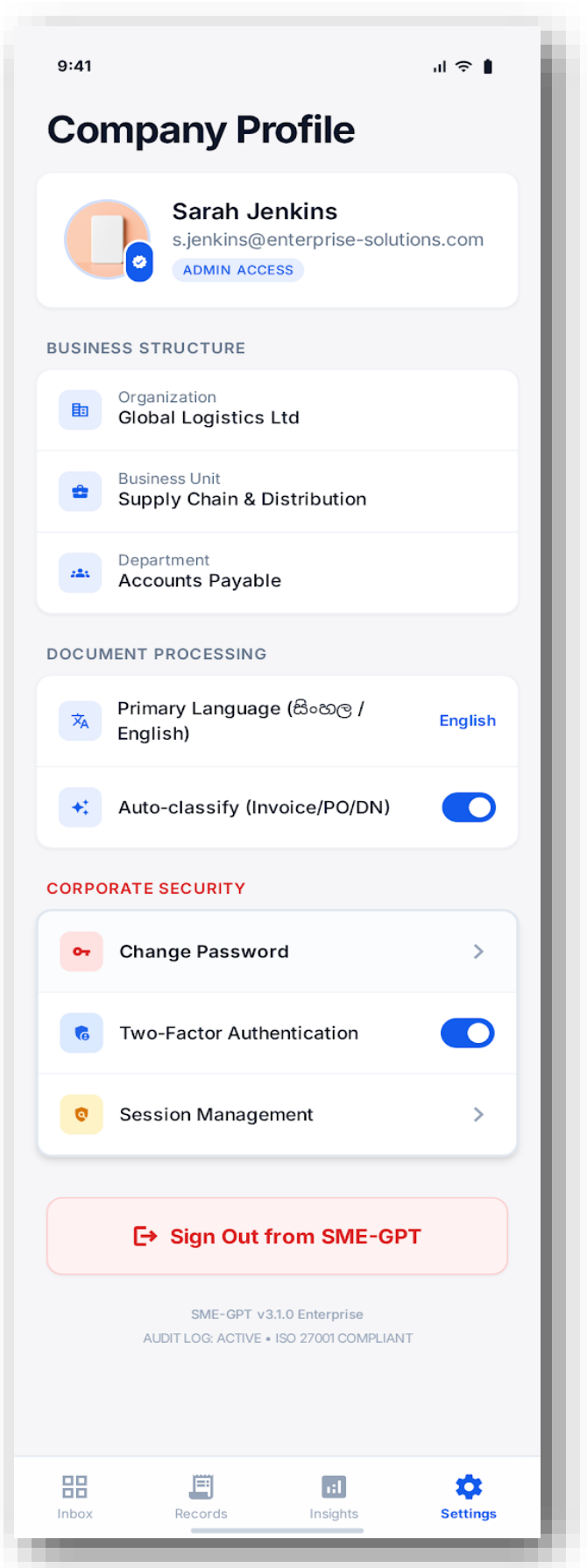
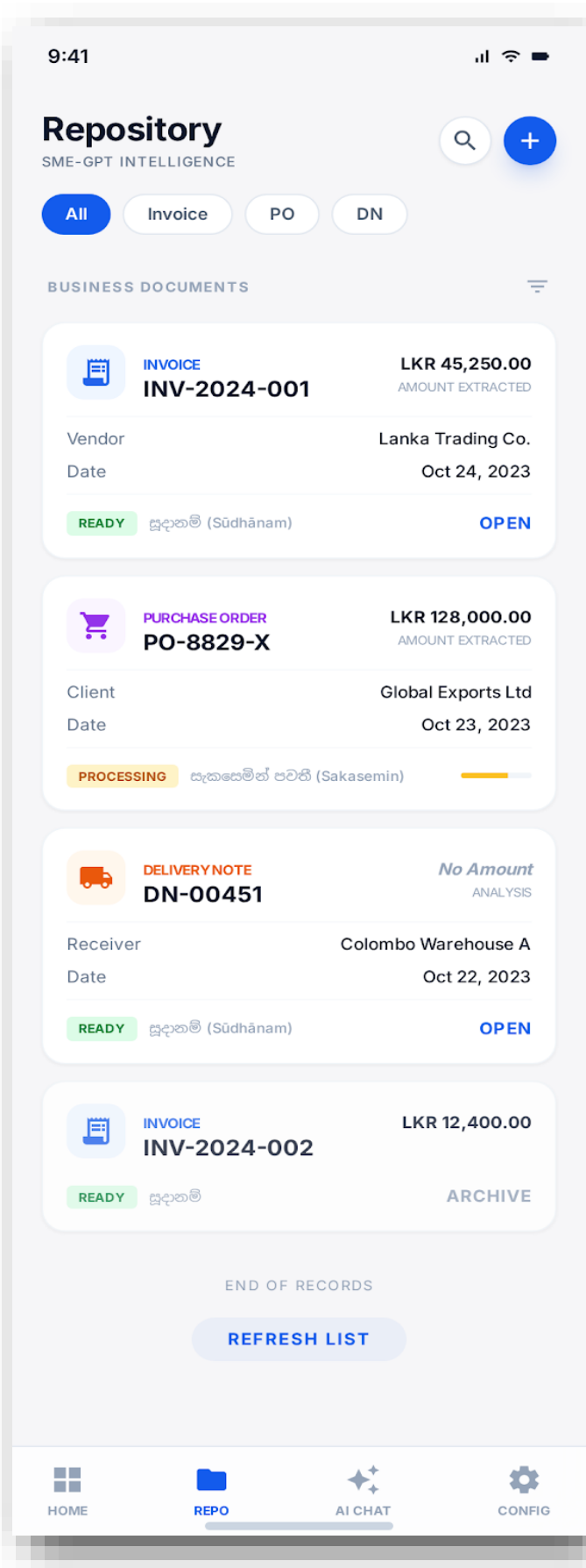
LINE 4

Database Record

Agreed Unit Price: Portland Cement

2,250.00

UI Design 6



UI Design 7

UI Design 8

The system shall provide a browser-based UI with the following main screens:

- Login/Logout
- Dashboard (documents + status)
- Upload screen (drag & drop), Chat/Q&A screen (question, answer, evidence, derivation steps)
- Document Viewer (PDF/image pages + overlay highlights)
- Extraction Results Panel (fields + confidence + click-to-source)
- Admin panel (user management, logs)

UI rules:

- Bilingual Sinhala/English toggle
- Clear success/failure messages
- Consistent button layouts (Upload, View, Delete, Ask)
- Standard error display area
- Responsive layout for mobile + desktop

4.2. Hardware Interfaces

No specialized hardware is required. The system must work with standard desktops/laptops/smart phones

4.3. Software Interfaces

The system interfaces with:

- OCR engine(s) (pluggable) via internal service calls/APIs
- Vector database for embeddings and retrieval
- Relational database for structured storage and user data
- PDF rendering library for page conversion
- LLM service (API or local model runtime)
- Data exchanged is primarily JSON objects including: text, bounding boxes, confidence, IDs, and metadata.

4.4. Communications Interfaces

- Communication shall use HTTPS (TLS)
- Token-based authentication for API access
- Secure upload channels and size limits
- Standard HTTP response codes and consistent error payload format
- Optional rate limiting to prevent abuse

5. Other Nonfunctional Requirements

5.1. Performance Requirements

NFR-01: OCR should process pages within seconds.

Why: Users expect fast document processing.

NFR-02: Query responses should be fast.

Why: Ensures good user experience.

NFR-03: System must handle large volumes of documents.

Why: SMEs may upload hundreds or thousands of invoices.

5.2. Safety Requirements

Although the system is not safety-critical (no physical harm), it must prevent “business harm” such as accidental deletion, incorrect financial totals being treated as facts, and data loss. Safeguards include:

- Confirmations before delete operations
- Read-only default mode for extracted values until user approves
- Warning labels for low-confidence OCR fields
- Backup strategy and recovery procedures
- Prevent the system from producing totals without showing evidence and calculation trace

5.3. Security Requirements

NFR-14: Must follow GDPR-like standards.

Why: Personal and financial data must be handled properly.

NFR-15: Audit logs must be permanent for 1 year.

Why: Necessary for audits and investigations.

Additionally should:

- Role-based access control (RBAC)
- Tenant isolation between SMEs
- Encryption in transit (TLS) and at rest (AES-256)
- Secure password hashing (e.g., bcrypt/argon2)
- Logging of authentication attempts and sensitive actions
- Least-privilege design (users only access what they need)

5.4. Software Quality Attributes

Reliability:

NFR-04: Automatic retry on OCR/layout errors.

Why: Prevents pipeline failure from minor issues.

NFR-05: Maintain 99% uptime.

Why: Reliability matters for daily business usage.

Usability:

NFR-06: UI should be simple and intuitive.

Why: Non-technical SME users must use it easily

NFR-07: Provide bilingual UI.

Why: Sinhala support is essential.

NFR-08: App should be installable on mobile and desktop devices.

Why: Improves access and adoption.

NFR-09: Responsive UI adapting to different screen sizes.

Why: SMEs use phones, tablets, desktops.

Maintainability:

NFR-10: System must be modular.

Why: OCR/layout/LLM will evolve; modules must be replaceable.

NFR-11: Easy to update or replace models.

Why: AI technology improves rapidly.

Portability:

NFR-12: System should run on different devices.

Why: SMEs may need local deployments for privacy.

NFR-13: Use Docker containers.

Why: Ensures consistent deployment environments.

6. Other Requirements

6.1. Data Requirements

6.1.1. Data Inputs

- PDFs/images of invoices, POs, delivery notes
- User queries in Sinhala/English

6.1.2. Data Outputs

- Structured extracted fields (JSON)
- Provenance metadata (page, bbox, confidence, raw text)
- Highlighted visual explanations
- LLM answers with derivation trace

6.1.3. Data Storage

- Document storage (original + processed images)
- OCR outputs
- Extraction metadata
- Vector embeddings
- Logs & audit trail

6.2. Internationalization

The UI and query handling shall support Sinhala and English. Date formats, currency formats, and number formatting should be consistent and configurable for local context.

6.3. Legal/Ethical

The system must handle financial documents responsibly. It should avoid exposing sensitive information across tenants and must ensure users understand AI confidence and limitations.

Appendix A: Glossary

OCR – Optical Character Recognition

NLP – Natural Language Processing

LLM – Large Language Model

RAG – Retrieval Augmented Generation

RBAC – Role-Based Access Control

TLS – Transport Layer Security

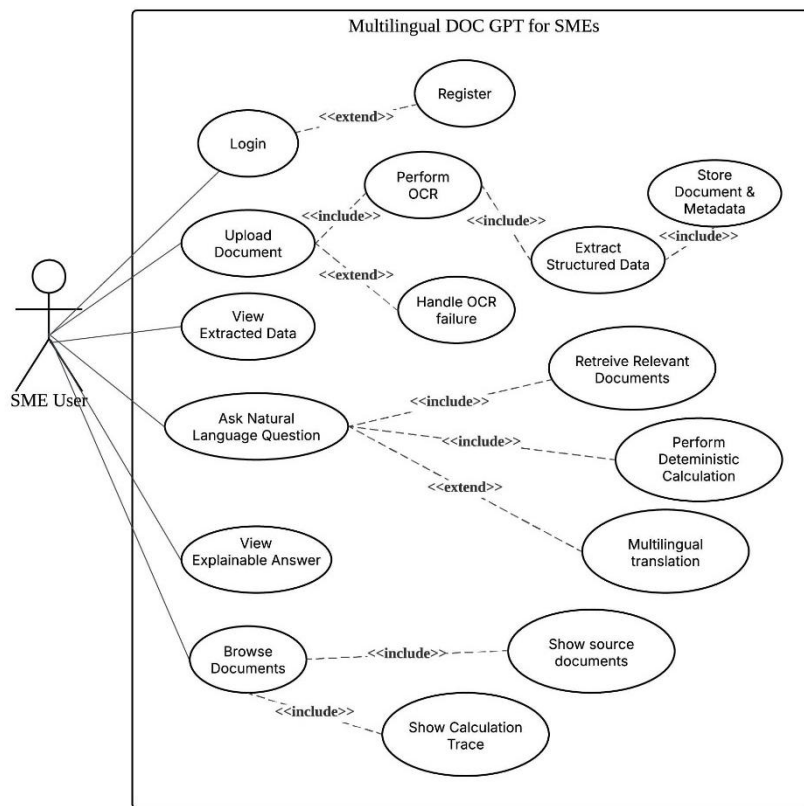
Provenance – Evidence trail showing where extracted/answered data came from

Embedding – Vector representation of text used for semantic retrieval

Vector DB – Database optimized for similarity search over embeddings

Appendix B: Analysis Models

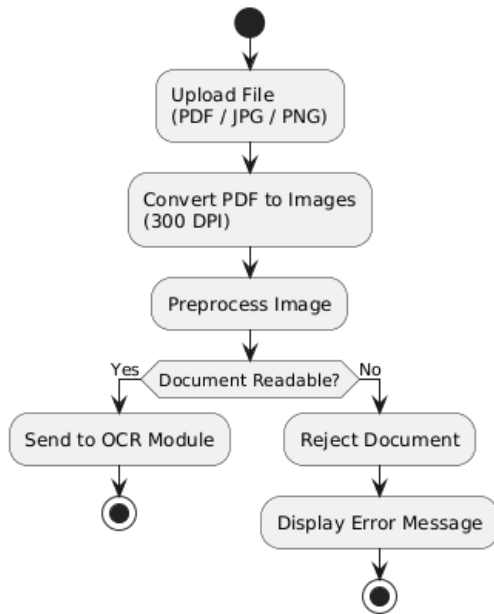
1. Use Case Diagram



Use Case Diagram 1

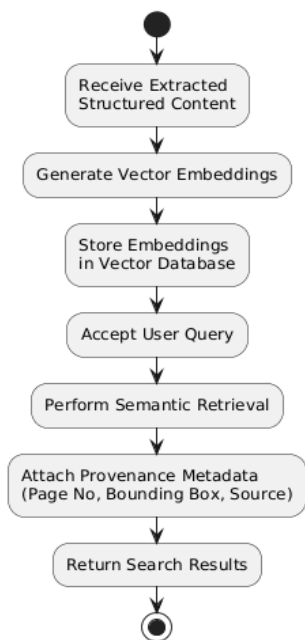
2. Activity Diagram

Upload File



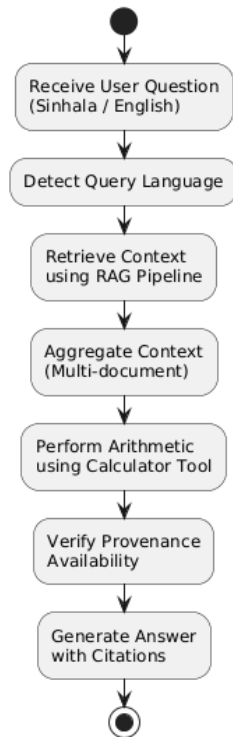
Activity Diagram 1 - Upload File

Indexing and Vector retrieval



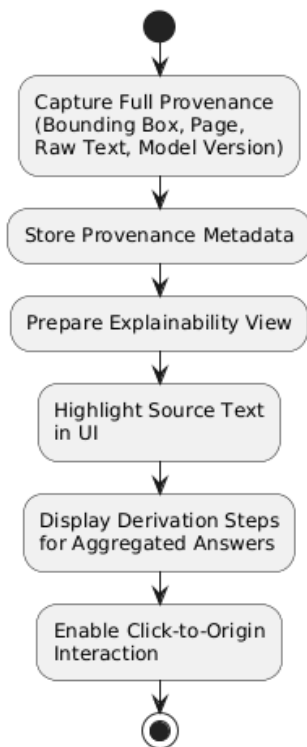
Activity Diagram 2 - Indexing and VR

LLM-Based Question Answering



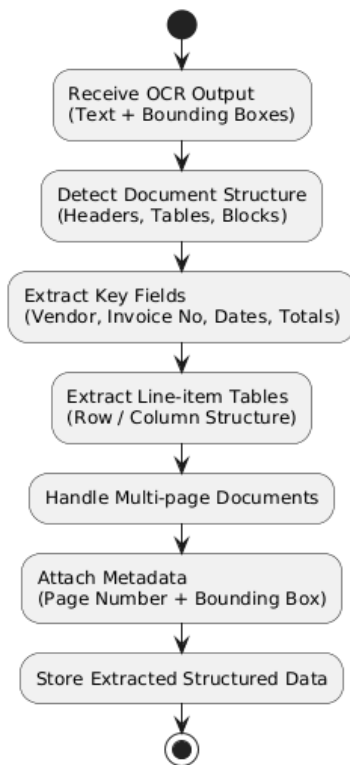
Activity Diagram 3 - LLM based QnA

Explainability and Provenance Tracking



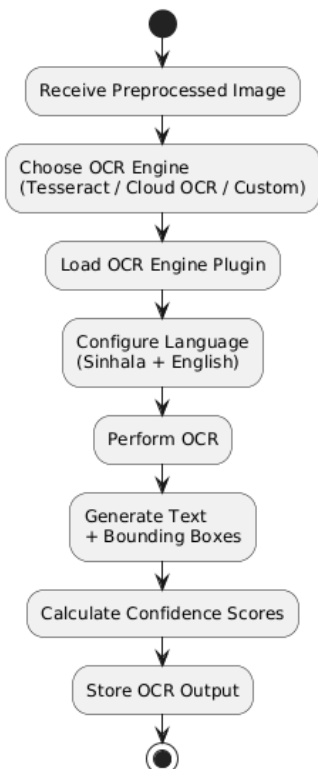
Activity Diagram 4 - Explainability

Layout Detection



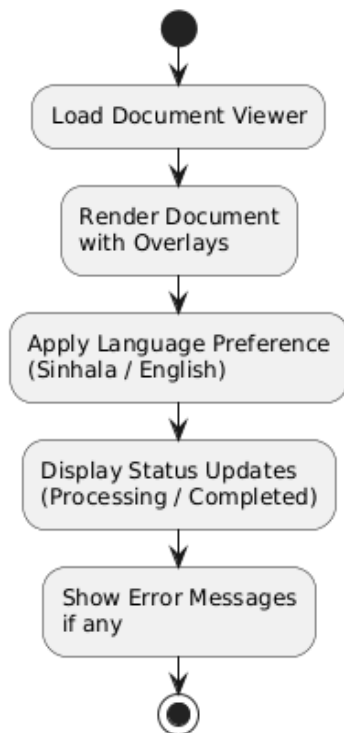
Activity Diagram 5 - Layout Detection

OCR processing (Sinhala + English)



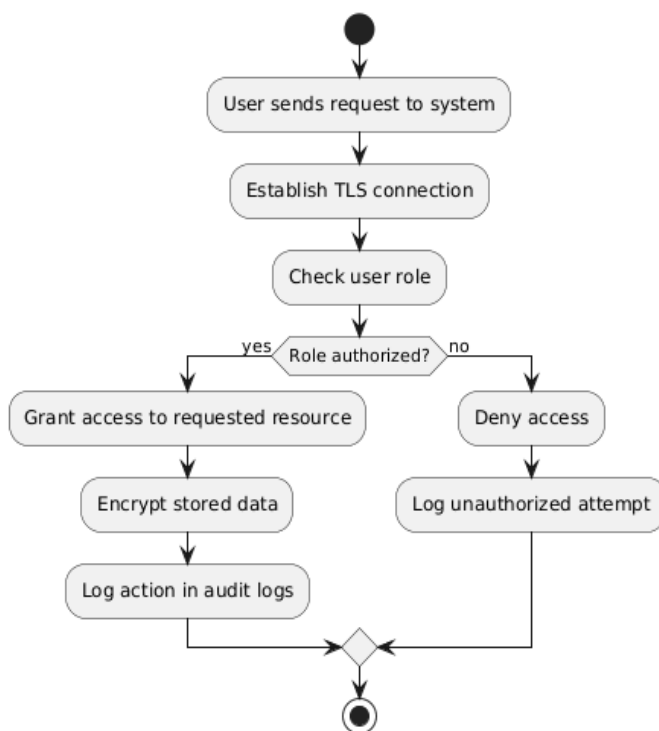
Activity Diagram 6 -OCR Processing

User Interface



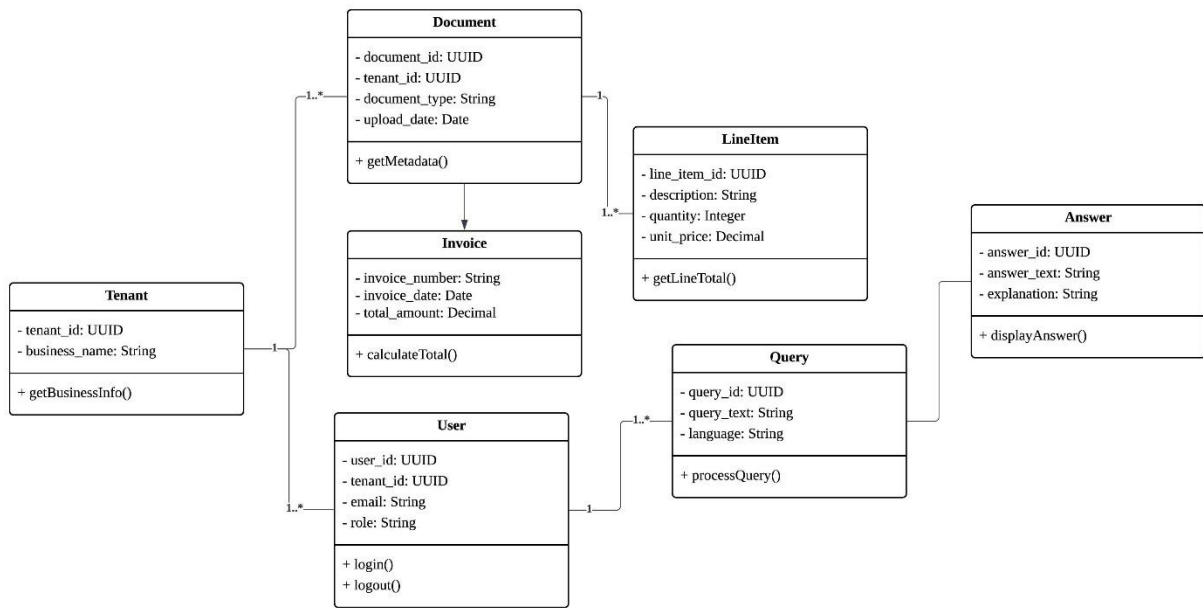
Activity Diagram 7 - User Interface

Security and Access Control



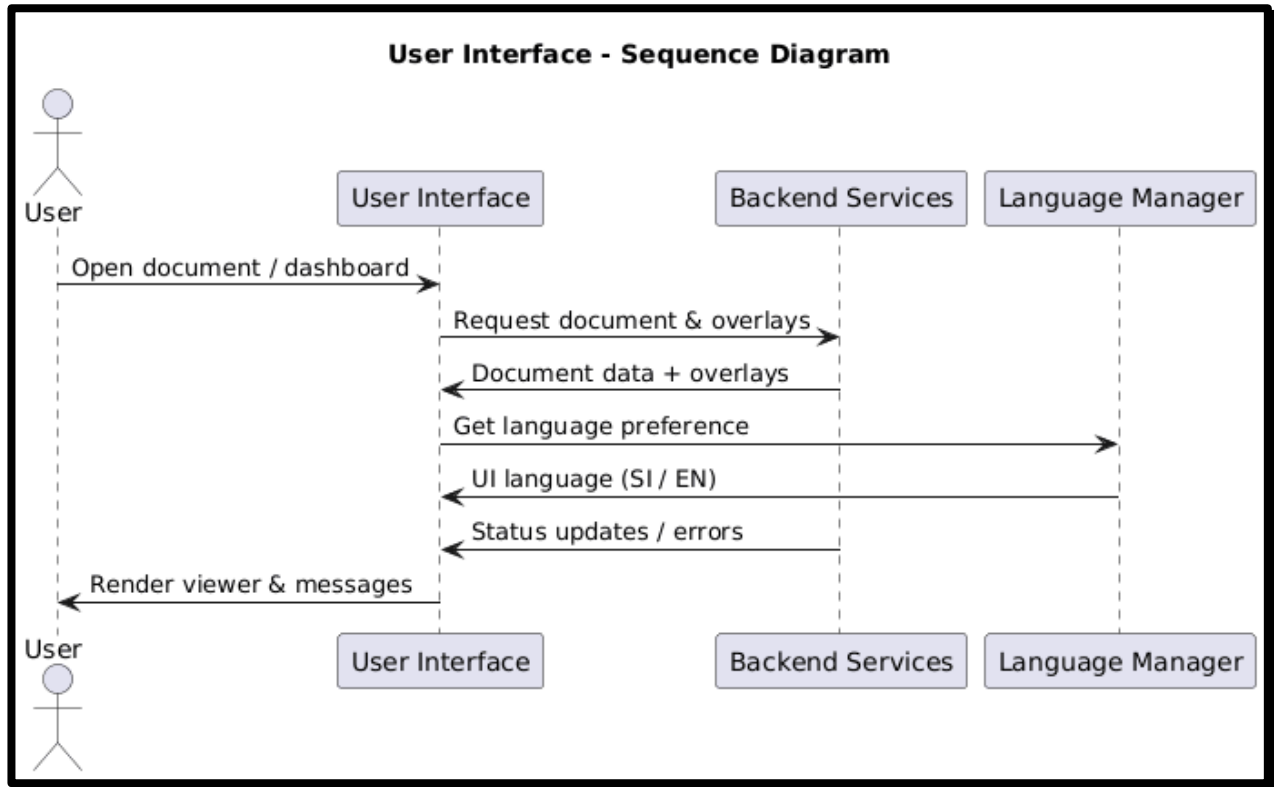
Activity Diagram 8 - Security and Access

3. Class Diagram

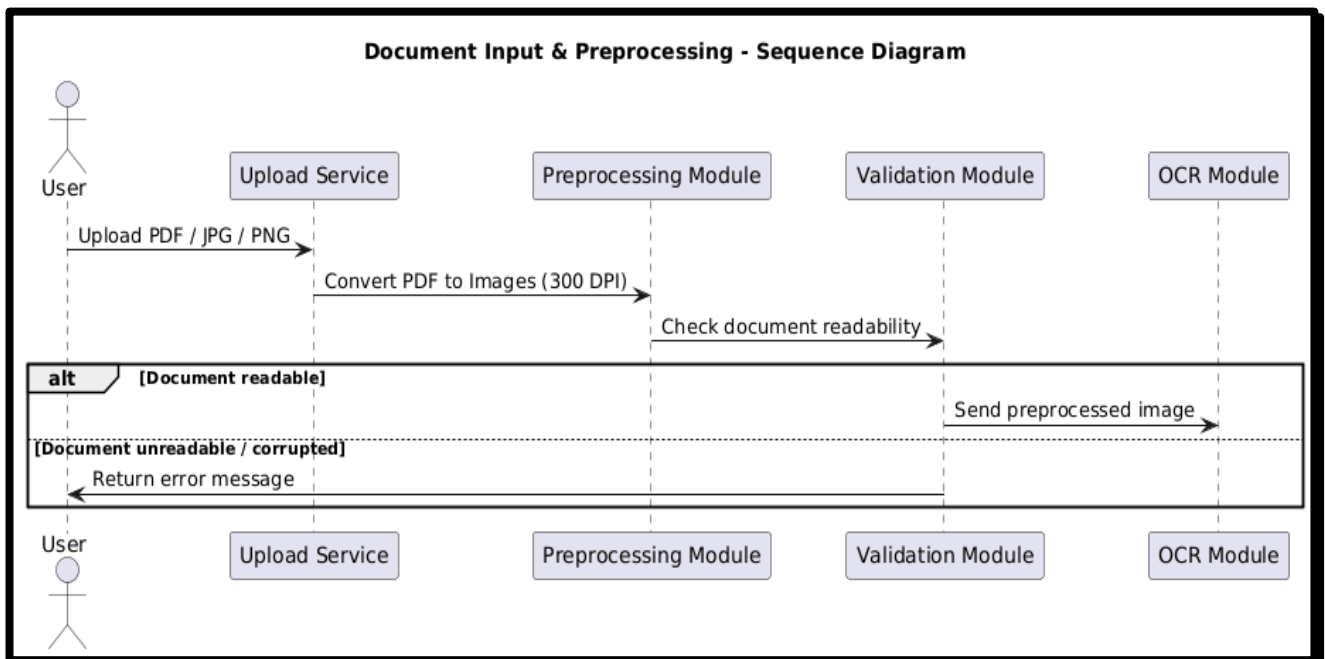


Class Diagram 1

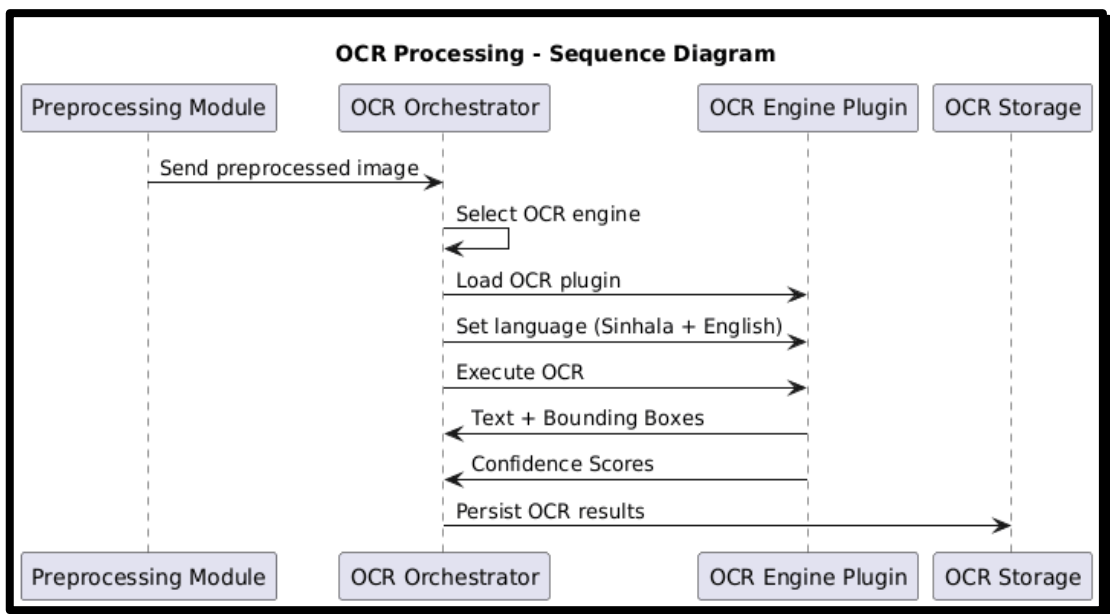
4. Sequence Diagram



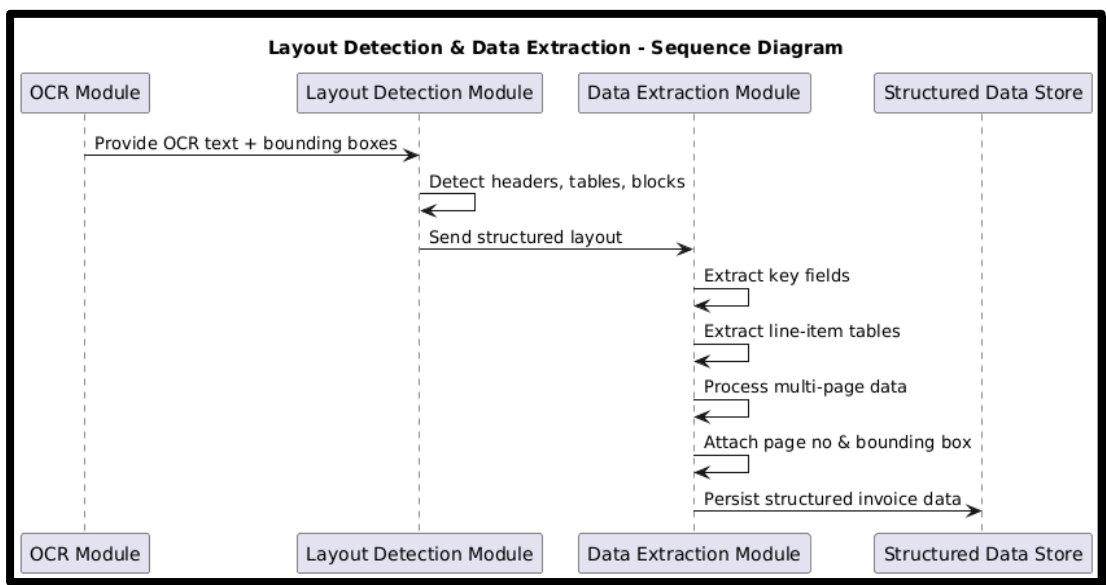
Sequence Diagram 1 – User Interface



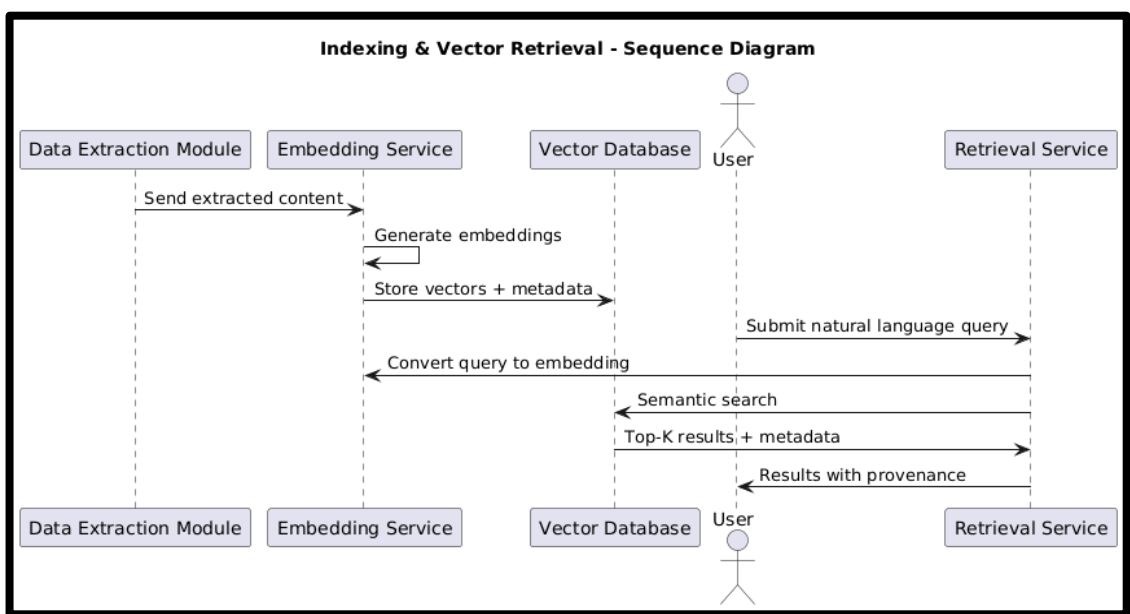
Sequence Diagram 2 – Document Upload



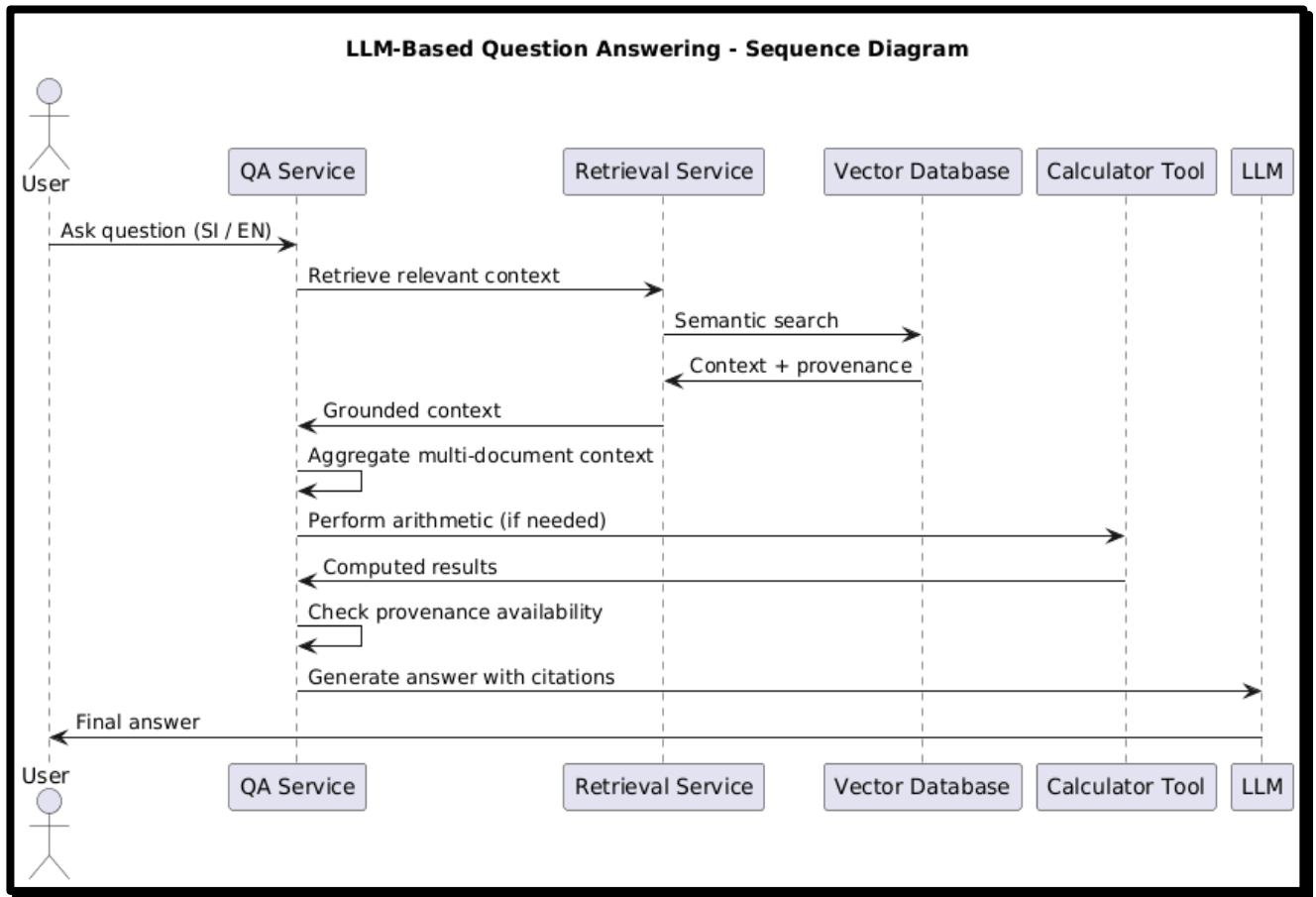
Sequence Diagram 3 – OCR Processing



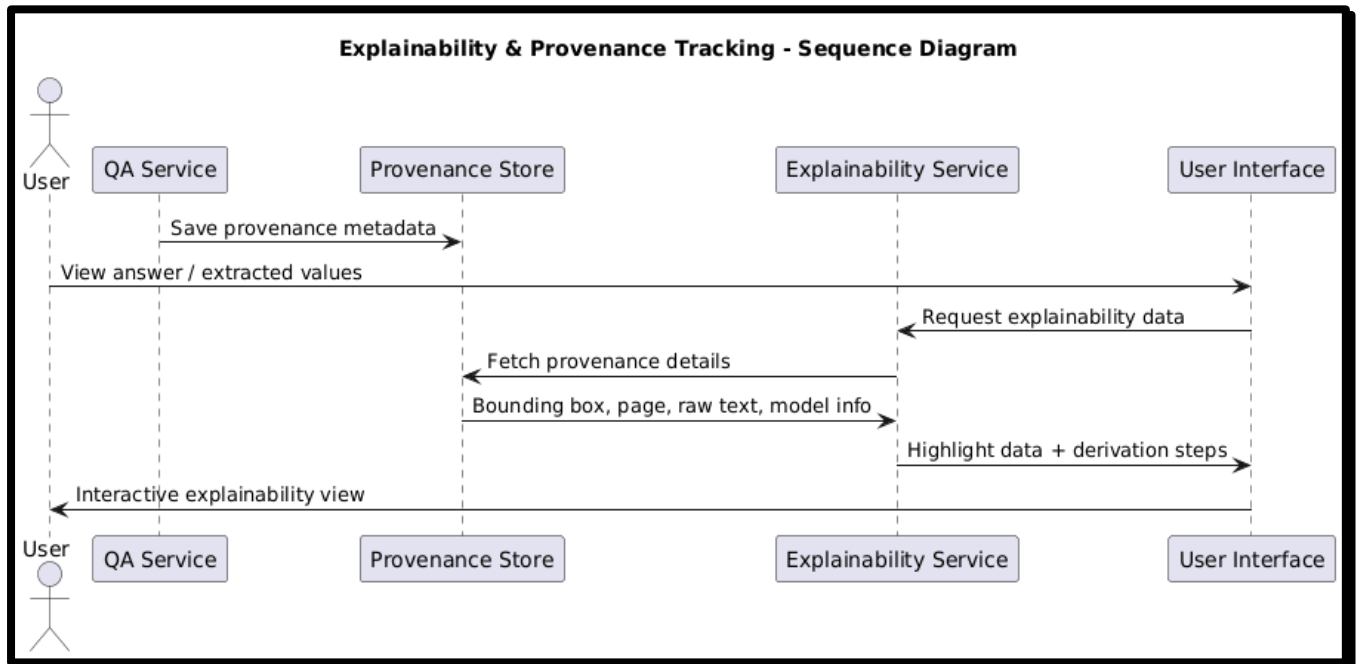
Sequence Diagram 4 – Layout Detection and Data extraction



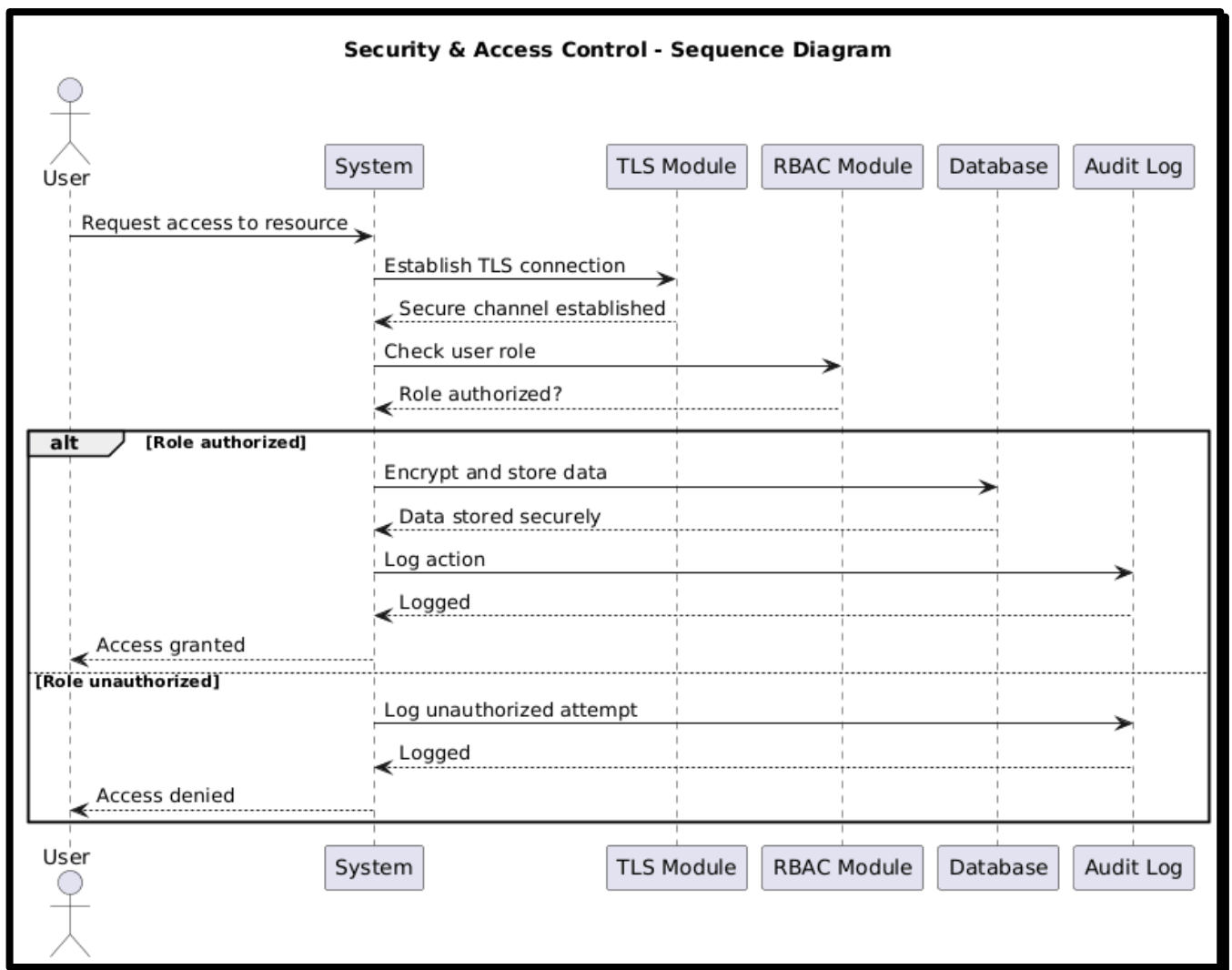
Sequence Diagram 5 – Indexing and Vector Retrieval



Sequence Diagram 6 – LLM based Question answering

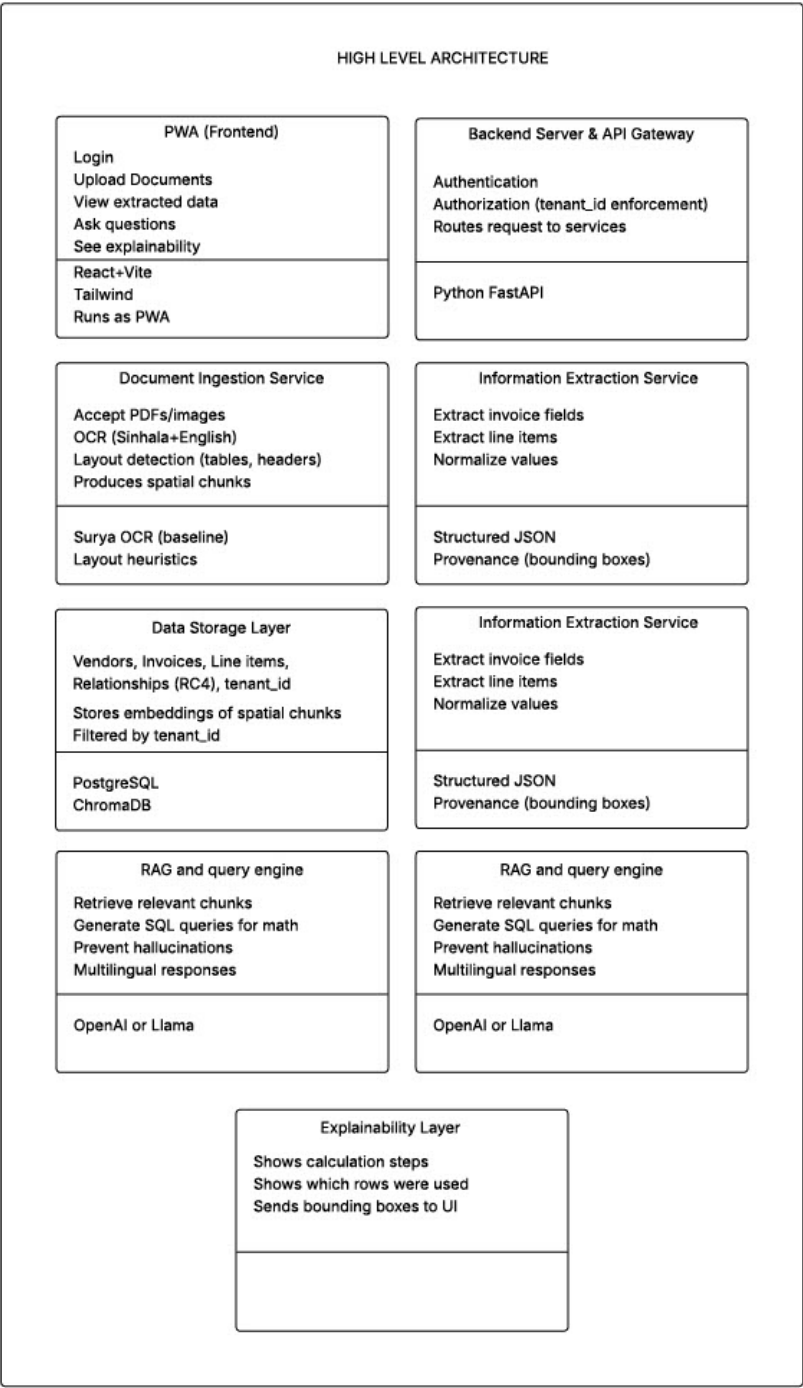


Sequence Diagram 7 – Explainability



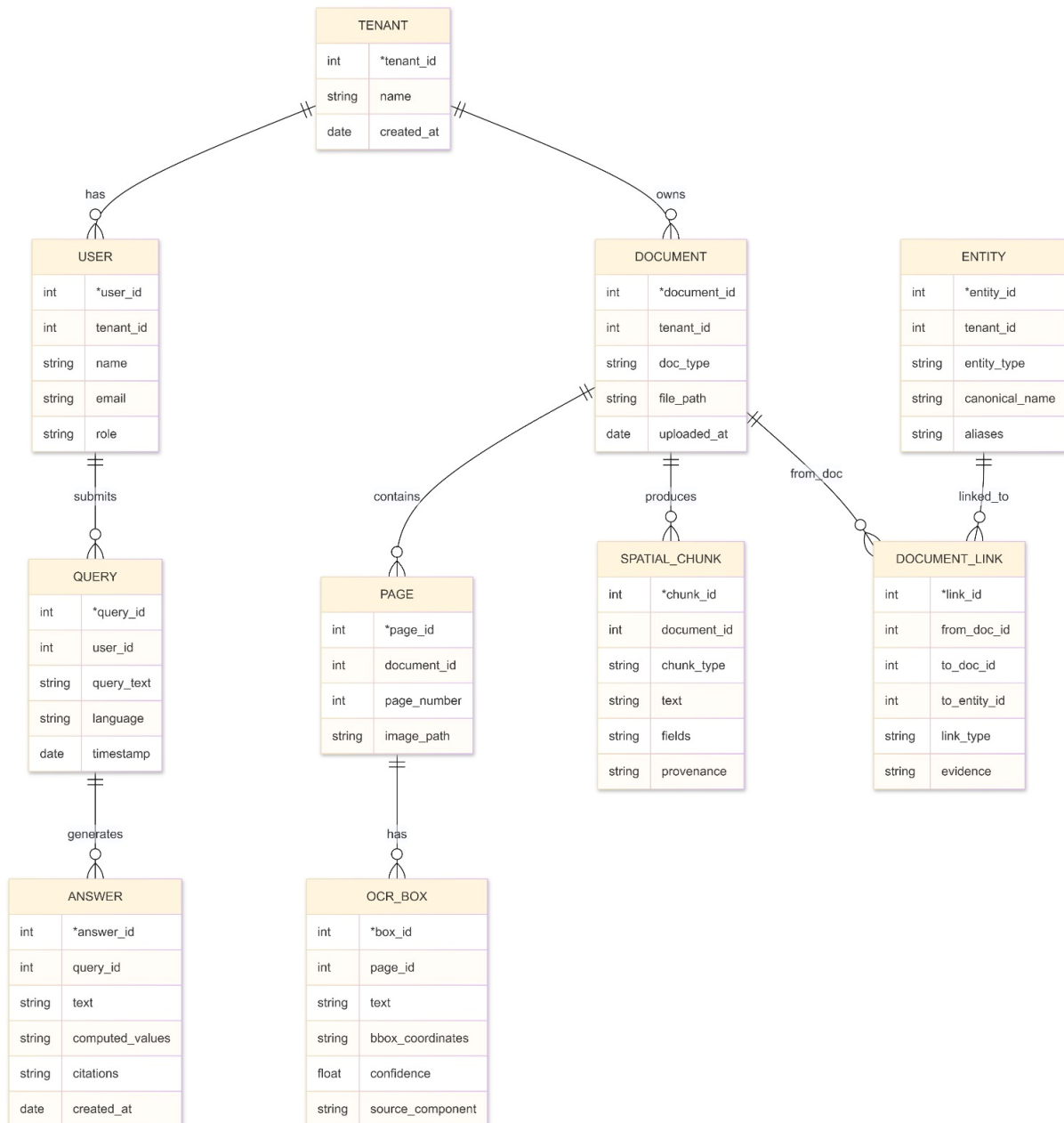
Sequence Diagram 8 – Security

5. High Level Architecture Diagram



High level Architecture diagram 1

6. ER Diagram



ER Diagram 1

Appendix C: Issues List

Open issues that require resolution during implementation and testing:

1. Sinhala OCR accuracy variability across fonts and low-quality images
2. Handling complex invoice layouts and inconsistent table structures
3. Defining best chunking strategy for embeddings (table rows vs sections)
4. Ensuring strict grounding and refusal behavior when evidence is insufficient
5. Performance optimization for large document volumes
6. Designing best UI experience for overlays and click-to-source explainability
7. Finalizing GDPR-like compliance approach and retention policies